

PQC Perf Modeling – Project Checklist

Section 1 – Finalize Dataset & Features

- ☐ 1. Create a single clean CSV (e.g., pqc_combined_clean.csv) with Kaggle + harness data.
- ☐ 2. Ensure consistent column names/units and add a dataset_source column.
- ☐ 3. Define the final feature list (categorical + numeric) and the target (log_latency).

Section 2 – Evaluation Setup

- ☐ 4. Choose a held-out deployment machine (e.g., AWS c6) and NEVER train on it.
- ☐ 5. Define train_machines = all machines except the held-out one.
- ☐ 6. Implement GroupKFold on machine_id using only train_machines.

Section 3 – Baseline Models

- ☐ 7. Baseline 0: global-mean model (predict mean log_latency).
- ☐ 8. Baseline 1: per-config mean model (e.g., by alg_param).
- ☐ 9. Baseline 2: Ridge regression with proper scaling/encoding.
- ☐ 10. For each baseline, record CV MAE/RMSE over GroupKFold.
- ☐ 11. Retrain each baseline on all train_machines and evaluate once on held-out machine.

Section 4 – Non-linear Models

- ☐ 12. Implement RandomForestRegressor with a small hyperparameter search using GroupKFold.
- ☐ 13. Optionally add one boosting model (e.g., Gradient Boosting or similar).
- ☐ 14. Log CV MAE/RMSE for each model and compare against Ridge and baselines.
- ☐ 15. Retrain best non-linear model(s) on all train_machines and evaluate on held-out machine.

Section 5 – Quantile Regression (p50 & p95)

- ☐ 16. Choose one model family for quantiles (e.g., linear QuantileRegressor or boosting).
- ☐ 17. Train models for tau = 0.5 (median) and tau = 0.95 (p95).
- ☐ 18. Evaluate quantile predictions on the held-out machine (calibration: % below p50/p95).
- ☐ 19. Create at least one plot of predicted p50/p95 vs actual latencies for a chosen config.

Section 6 – Robustness & Feature Importance

- ☐ 20. Compute and plot Ridge coefficients (standardized) for key features.
- ☐ 21. Compute and plot Random Forest feature_importances_.
- ☐ 22. Compare models with vs without hardware features (cpu_ghz, aesni, avx2, neon, containerized).
- ☐ 23. Summarize per-machine MAE/RMSE for Ridge vs Random Forest.
- ☐ 24. Plot error vs concurrency and/or message_bytes to see where models struggle.

Section 7 – EDA & Plots for the Report

- ☐ 25. Create box/violin plots of log_latency by alg_family (optionally split by dataset_source).
- ☐ 26. Plot log_latency vs message_bytes (facet by algorithm or machine).
- ☐ 27. Summarize mean/median latency by machine_id to show hardware heterogeneity.

Section 8 – Writeup & Alignment with Proposal

- ☐ 28. Describe data sources, cleaning steps, and final feature set.
- ☐ 29. Explain the evaluation protocol (GroupKFold + held-out deployment machine).
- ☐ 30. Present results for baselines, Ridge, and Random Forest (CV + held-out machine).

- [] 31. Present quantile regression results and plots for p50/p95 latency.
- [] 32. Discuss robustness across machines and importance of hardware/workload features.
- [] 33. Tie the findings back explicitly to the goals in the original project proposal.